MIDTERM PROJECT – COMPUTER VISION (INS3155)

Traditional Image Processing for Filtering, 3D Reconstruction, and Image Stitching

Release Date: October 16, 2026

Submission Platform: Microsoft Teams (MS Teams)

Submission Deadline: October 21, 2026 (23:59 GMT+7)

Assessment Type: Individual Project (100 Points)

1. General Instructions & Submission Policy

- **Individual Work:** This is strictly an individual project. Collaboration, code sharing, or plagiarism is not allowed and will result in a zero grade.
- **Methodology Restriction:** You are **only** allowed to use **traditional image processing and classical computer vision methods**. Do **not** use any deep learning models or pretrained neural networks.
- **Allowed Libraries:** You may use OpenCV, NumPy, SciPy, Matplotlib, Scikit-image, Open3D, or other non-learning libraries.
- **Submission Requirements on MS Teams:** Submit all deliverables directly to the assignment tab on **Microsoft Teams** by **October 21, 2026**:
  a) **Project Report (PDF):** Named as `StudentID_FullName_Report.pdf`.
  b) **Source Code & Results (ZIP or GitHub Link):** Named as `StudentID_FullName_Code.zip` containing source code, sample test images, generated visual outputs, and a `README.md` explaining how to execute the scripts.

2. Project Objectives

You will implement, evaluate, and systematically compare classical computer vision techniques across three core tasks:

1) **Part A:** Image filtering, noise reduction, and edge sharpening.
2) **Part B:** 3D reconstruction and epipolar geometry from stereo images.
3) **Part C:** Multi-view feature matching and panoramic image stitching.

For each part, you must conduct **multi-method comparisons** supported by both **quantitative metrics** (accuracy, error rates, runtime) and **qualitative visual evaluation**.

3. Detailed Task Specifications

Part A – Image Filtering and Enhancement (20 Points)

Implementation Tasks:

- Synthesize noisy test images from clean images using at least two noise models: **Gaussian noise** and **Salt-and-Pepper noise**.
- Apply and compare four traditional spatial filters across multiple kernel sizes (e.g., $3 \times 3, 5 \times 5, 7 \times 7$):
  - **Mean Filter** (Box filter)
  - **Gaussian Filter** (with different standard deviations σ)
  - **Median Filter**
  - **Laplacian Sharpening** (for detail and edge enhancement)

Deliverables & Evaluation:

- **Visual Results:** Side-by-side grids showing original, noisy, denoised, and sharpened images.
- **Quantitative Evaluation:** A comparative table reporting **PSNR (dB)**, **SSIM**, **MSE**, and **Execution Time (ms)** for each filter and parameter setting across both noise types.
- **Qualitative Evaluation:** Visual discussion comparing noise removal capability vs. edge preservation, over-smoothing effects, and halo/ringing artifacts.

Part B – 3D Reconstruction from Stereo Images (25 Points)

Implementation Tasks:

- Detect and match stereo correspondences to estimate the **Fundamental Matrix (F)** via RANSAC, and draw **epipolar lines** on both left and right images.
- Compute dense **disparity maps** by implementing and comparing **both** methods:
  - **Block Matching (BM)**
  - **Semi-Global Block Matching (SGBM)**
- Evaluate different stereo parameter settings (e.g., varying `blockSize` and `numDisparities`).
- Reconstruct the **3D point cloud** from the disparity map and camera parameters.

Deliverables & Evaluation:

- **Visual Results:** Epipolar lines on stereo pairs, colorized disparity maps (BM vs. SGBM), and multi-view screenshots of the 3D point cloud.
- **Quantitative Evaluation:** Compare BM vs. SGBM and parameter settings using:
  - **Valid Disparity Ratio (%):** Percentage of valid (non-invalid/non-occluded) pixels.
  - **Mean Epipolar / Sampson Error (px):** Distance of matched points to epipolar lines.
  - **RMSE / Bad-Pixel Rate** (if ground truth is available) and **Runtime (ms)**.
- **Qualitative Evaluation:** Visual analysis of disparity smoothness, object boundary sharpness, artifacts in textureless/occluded areas, and 3D geometry accuracy.

Part C – Multi-View Image Stitching (25 Points)

Implementation Tasks:

- Build an end-to-end stitching pipeline for **four (4) overlapping images**:

1. **Feature Extraction:** Implement and compare at least **two** detectors/descriptors (e.g., **SIFT vs. ORB**, or **SURF / AKAZE**).
2. **Feature Matching:** Match keypoints across overlapping views using KNN matching and Lowe's ratio test.
3. **Homography Estimation:** Estimate 3×3 homography matrices using **RANSAC** across different thresholds.
4. **Warping & Blending:** Warp images into a common coordinate frame and blend overlapping regions (e.g., feathering/alpha blending or multi-band blending) into a single panorama.

Deliverables & Evaluation:

- **Visual Results:** Keypoint visualizations, matched inlier lines, and final stitched panoramas for each compared feature extractor.
- **Quantitative Evaluation:** A comparison table across feature detectors (e.g., SIFT vs. ORB) reporting:
  - **Number of Detected Keypoints** and **Initial Matches**.
  - **Number of Inliers** and **Inlier Ratio (%)** after RANSAC.
  - **Mean Reprojection Error (RMSE)** of inliers and **Total Stitching Time (ms)**.
- **Qualitative Evaluation:** Visual inspection of seam visibility, geometric alignment, ghosting artifacts, illumination transitions, and perspective distortion.

4. Comparative Analysis & Report Structure

Submit a clearly written PDF report structured into the following sections:

1. **Introduction & Motivation:** Overview of objectives and real-world applications.
2. **Methodology:** Mathematical equations, algorithmic workflows, and block diagrams for Parts A, B, and C.
3. **Implementation & Experimental Setup:** Dataset details, environment setup, and parameter choices.
4. **Comparative Analysis (Required – 15 Points):** Comprehensive synthesis comparing at least two methods/parameter configurations per part:
   - **Quantitative Analysis:** Structured tables and charts summarizing all metrics (PSNR, SSIM, Valid Pixels, Inlier Ratio, RMSE, Runtime).
   - **Qualitative Analysis:** Side-by-side visual comparisons highlighting strengths, weaknesses, and failure cases.
5. **Conclusion & Future Work:** Key takeaways and potential algorithmic improvements.

5. Grading Rubric (100 Points)

Bonus Points (+5 to +10 Points): Awarded for capturing and evaluating your own custom stereo/multi-view photographs, implementing advanced blending (e.g., Laplacian pyramid blending), or conducting insightful parameter sensitivity studies.

Component	Evaluation Criteria	Points
Part A: Image Filtering	Implementation of 4 filters on multiple noise models	**20**
Part B: 3D Reconstruction	Fundamental matrix, epipolar lines, BM vs. SGBM disparity maps, and 3D point cloud	**25**
Part C: Image Stitching	Multi-detector comparison (e.g., SIFT vs. ORB), RANSAC homography, 4-image panorama	**25**
Comparative Analysis	Rigorous **quantitative** metrics (PSNR/SSIM, inliers, RMSE, runtime) & **qualitative** analysis	**15**
Report Quality	Clarity, mathematical explanations, formatting, figures/tables organization	**10**
Code Quality & Submission	Clean, reproducible code and on-time submission on **MS Teams** by **21/10/2026**	**5**
Total Score		**100**